

# Nutrition Assistant Project

## Phase 5: Project Development

### Introduction

The Project Development phase is the implementation stage where the Nutrition Assistant application is built according to the approved design and requirements. During this phase, developers create the frontend and backend modules, integrate APIs, connect the database, and ensure that all project components function together efficiently.

### Codebase Structure

The project is organized into two major directories:

**Client Folder:** Contains the React.js frontend, reusable UI components, routing, state management, pages, stylesheets, and API service files.

**Server Folder:** Contains the Node.js and Express backend, REST APIs, authentication logic, middleware, controllers, database models, configuration files, and business logic.

This modular folder structure improves readability, scalability, and maintainability.

### Development Standards

Git Version Control is used to manage source code. Branching Strategy includes **main**, **dev**, and **feature/branch-name**. Meaningful commit messages are maintained for every update. Consistent naming conventions are followed for files, folders, variables, and functions. Reusable components are created to reduce code duplication. Code formatting and documentation are maintained for better collaboration.

### Development Workflow

The development process begins with frontend UI implementation followed by backend API development. After both modules are completed, API integration is performed to enable communication between the client and server. The application is then tested locally before deployment.

Both frontend and backend servers can be executed simultaneously using development commands such as **npm install** followed by **npm run dev**. Environment variables are stored securely using configuration files.

### Key Modules Developed

User Registration and Login User Profile Management Food Search and Nutrition Database Meal Logging System Daily Calorie and Macronutrient Calculator Dashboard with Charts and Progress Reports Water Intake and Reminder Module Authentication and Session Management

### Tools and Technologies

Frontend: React.js, Vite, HTML5, CSS3, Tailwind CSS, JavaScript.

Backend: Node.js, Express.js.

Database: MongoDB.

Version Control: Git and GitHub.

API Testing: Postman.

Code Editor: Visual Studio Code.

### Conclusion

The Project Development phase converts the project design into a working application. By following coding standards, modular architecture, and version control practices, the Nutrition Assistant

becomes easier to maintain, test, and enhance in future releases.